Michele Tarquini

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Leinweber Graduate Fellow in Theoretical Physics | Caltech

Education

Caltech

2025 – present Math-AI group

2022 – present Ph.D. in Theoretical Physics — Expected Graduation: 01/2028
Advisors: C. Cheung and S. Gukov.

Sapienza University of Rome

2020 – 2022 M.Sc. in Theoretical Physics — Final grade: 110/110 *cum laude*.

2017 – 2020 B.Sc. in Physics — Final grade: 110/110 *cum laude*.

Publications

Machine Learning for Mathematics

[1] *The Two-Hump Problem: Bridging the Difficulty Gap in Mathematical Reinforcement Learning.*
L. Fagan*, **M. Tarquini***, A. Shehper*, M. Manko, A. Gruen, C. Huang, G. Butbaia, D. Passaro, S. Gukov.
ICML (2026). arXiv:2606.21611; code and datasets. *Equal contribution.

[2] *New Snake-in-the-Box Records via Snakepit Surgery and Learned Construction.*
P. Orland*, L. Fagan*, **M. Tarquini***, D. Passaro, M. Manko, E. Heyes, A. Gruen, G. Butbaia, J. Tan, S. Gukov.
Preprint (2026). arXiv:2607.15270. *Equal contribution.

[3] *New Records for the Hadamard Maximal Determinant Problem.*
G. Butbaia, J. Tan, P. Vipulanandan, X. Huang, T. Saunders-A'Court, L. Fagan, D. Passaro, **M. Tarquini**, S. Gukov.
Preprint (2026). arXiv:2608.22518.

[4] *Hierarchical Reinforcement Learning for Sparse-Reward Search in Commutative Algebra.*
G. Butbaia, P. Orland, C. Huang, D. Passaro, L. Fagan, **M. Tarquini**, H. Dao, D. Eisenbud, A. Shehper, S. Gukov.
ICML (2026). arXiv:2606.22922.

Theoretical Physics

[5] *The Equivalence Principle at High Energies Completes the Spectrum.*
F. Calisto, C. Cheung, G. N. Remmen, F. Sciotti, **M. Tarquini**.
Preprint (2026). arXiv:2605.20319.

[6] *Inverse Problem in Effective Field Theory.*
F. Calisto, C. Cheung, G. N. Remmen, F. Sciotti, **M. Tarquini**.
Phys. Rev. Lett. **137**, 071601 (2026). arXiv:2604.15423.

[7] *Completeness from Gravitational Scattering.*
F. Calisto, C. Cheung, G. N. Remmen, F. Sciotti, **M. Tarquini**.
Phys. Rev. D **113**, 106008 (2026). arXiv:2512.11955.

[8] *Strings from Almost Nothing.*
C. Cheung, G. N. Remmen, F. Sciotti, **M. Tarquini**.
Phys. Rev. Lett. **136**, 251601 (2026). arXiv:2508.09246.

[9] *The Role of the Pion in the Lineshape of the $X(3872)$.*
A. Esposito, D. Germani, A. Glioti, A. D. Polosa, R. Rattazzi, **M. Tarquini**.
Phys. Lett. B **847**, 138285 (2023). arXiv:2307.11400.

Honors & Awards

2026 **Leinweber Graduate Fellowship**

2026 **John S. Stemple Memorial Prize**
For excellence in the Ph.D. candidacy exam.

Conferences, Workshops & Seminars

2026 **Machine Learning:** ICML; AIM Workshop on Number Theory and AI.

2025–2026 **Physics:** Pheno 2025; California Amplitudes Meeting; Caltech and UCLA physics seminars.

Teaching

2022–present Teaching Assistant: Waves, Quantum Mechanics, Quantum Field Theory.